

Toxicity Detection for Generative AI Systems: A Survey of Evidence and Governance Validity

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Toxicity detection is essential for limiting harmful exposure without suppressing legitimate expression. Generative systems complicate detection through interaction, multimodal composition, and adaptive access. A structured integrative review of 183 works examines when detection evidence supports governance decisions. SCOPE organizes evidence by safety policy, control channel, observation unit, probability regime, and enforcement action. The synthesis identifies channel substitution, repeated-access amplification, and judge dependence as recurring limits on evidence transfer. Worked cases expose unsupported extensions and specify necessary comparisons. Governance claims require evidence matched to deployment conditions and intervention consequences. SCOPE makes these gaps inspectable but requires empirical validation.

CCS Concepts: • **General and reference** → **Surveys and overviews**; • **Computing methodologies** → **Natural language processing**; **Computer vision**; • **Security and privacy** → *Human and societal aspects of security and privacy*; • **Human-centered computing** → *Empirical studies in HCI*.

Additional Key Words and Phrases: toxicity detection for generative AI, content moderation, generative AI safety, guard models, multimodal safety, text-to-image safety, model components, calibration, governance

1 Introduction

Generative AI systems participate in communication by producing responses and images that people may encounter, reuse, and share. When these outputs express harassment, identity attacks, or threats, the system becomes a source of harmful content as well as a means of processing it. Toxicity detection is therefore necessary for identifying harmful exposure, directing moderation, and evaluating whether safeguards work. These decisions also affect legitimate

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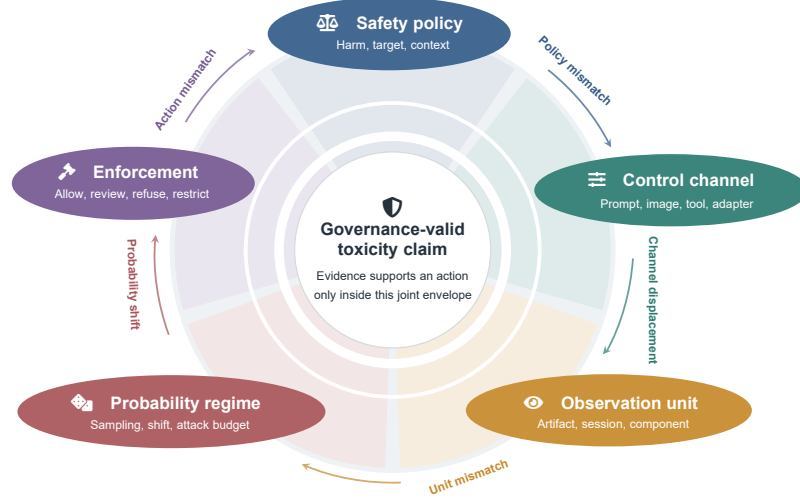


Fig. 1. The SCOPE validity envelope. A toxicity claim is supported only for the joint policy, channel, observation, probability, and action conditions represented by its evidence. The outer ring shows recurring failure modes when one coordinate changes.

expression: toxicity depends on the affected target, conversational and cultural context, platform policy, and annotator perspective, rather than a stable lexical property [18, 33, 124, 129, 131]. Missed abuse and unwarranted restrictions are both consequential. Even a detector with high aggregate accuracy can incorrectly flag identity terms, minority dialects, quotation, counterspeech, or reclaimed language, making participation more difficult for the people a moderation policy is intended to protect.

Generation makes this dual requirement harder to satisfy because input screening does not establish the safety of the resulting behavior. A benign prompt can yield harm, a nominal refusal can reveal actionable details, and individually harmless turns can compose into a harmful trajectory [35, 45, 127]. Detection must consequently account for the relationship between requests, outputs, and interaction history. Images, retrieval, tools, fine-tuning, learned tokens, and adapters add routes through which behavior can change, including routes that a prompt-only check cannot observe. Seeds, decoding, versions, repeated attempts, and adaptive attacks also determine which harmful outcomes become reachable [62, 80, 94]. These conditions matter to ordinary evaluation as well as attack testing: a score on one output, a rate across repeated generations, and a judgment about a complete conversation answer different questions about exposure and intervention.

The practical challenge is thus to determine what a detection result can support when it is used to allow content, refuse a request, route a case for review, or restrict a component. Comparing benchmark scores alone leaves this evidence-to-decision gap unresolved. This survey asks *when toxicity evidence justifies an action in a generative system*. We call this bounded relation *governance validity*. Its five coordinates form SCOPE: Safety policy, Control channel and carrier, Observation and adjudication unit, Probability regime, and Enforcement action. Figure 1 organizes these conditions. The framework assesses support for a stated claim, not the legitimacy of a policy or the safety of an entire system.

Four overlapping changes make this gap a common problem across the literature. They connect the need to recognize harmful content with the additional evidence required to govern generated behavior and reusable components.

- (1) **Labels become policies.** Binary targets suppress distinctions among hate, harassment, sexual exploitation, threats, and benign-sensitive discourse. Guard taxonomies and deployment rules remain only partly compatible [3, 45].
- (2) **Instances become behavior distributions.** Sampling, interaction, attacks, and model modification turn toxicity into an event probability conditioned on system state and access [35, 94, 115].
- (3) **Visible outputs acquire hidden carriers.** Embeddings, adapters, checkpoints, and edited states can encode, suppress, or reactivate harmful behavior after composition [29, 80, 89, 163].
- (4) **Accuracy becomes a decision problem.** Misses expose people to harm; false alarms suppress legitimate sensitive speech. Calibration, abstention, severity, reversibility, and review capacity determine whether a score improves governance [1, 121, 140].

1.1 Scope and Boundary

We use *toxicity* for communication that attacks, degrades, threatens, sexually exploits, or otherwise inflicts targeted social harm on people or groups. The scope has three layers. *Core evidence* retains a policy-defined toxicity endpoint. *Conditionally relevant safety evidence*, including studies of not-safe-for-work (NSFW) content or violence, enters quantitative toxicity synthesis only when its categories can be mapped to that endpoint. *Mechanism evidence*, including generic backdoors, model editing, and tool injection, explains how harm can be controlled or observed but supplies no toxicity rate by itself. Self-harm, illegal assistance, cyber abuse, privacy, copyright, provenance, and factuality are not interchangeable with toxicity; unrelated outcomes remain outside the synthesis. Text and image-based systems, including text-to-image (T2I) generators, are the principal settings; audio, video, and embodied interaction are not systematically covered.

Table 1 compares the scope and organizing questions of related surveys. Earlier work organizes hate-speech constructs and classifiers, compares dataset ontologies, or analyzes bias [27, 33, 132]. The AIGC review [151] treats non-toxicity as part of a wider risk taxonomy. Instead, we examine whether a *claim* survives a change in policy, governed object, access, or action. Evaluation studies such as SafetyBench, Do-Not-Answer, and MMSafeAware provide complementary benchmark evidence [152, 153, 179].

1.2 Research Questions and Contributions

The synthesis answers four questions. **RQ1** asks how toxicity is operationalized and which observation units and probability regimes are used to measure it. **RQ2** asks how evidence changes as the governed object moves from content to interactive, multimodal, and component-mediated behavior. **RQ3** asks which validity failures recur across these settings and why apparently compatible scores do not transfer. **RQ4** asks which evaluation and decision practices can support bounded, falsifiable governance claims.

We address these questions through a structured integrative review of 183 works. The synthesis contributes a five-coordinate support profile, four worked applications that identify missing transfer tests, and a cross-domain analysis of channel substitution, repeated-access amplification, and judge dependence. It shows why evidence about a label, a sampled outcome, or a static component cannot by itself establish conversation safety or justify a stronger enforcement action. These findings motivate a proposed assurance loop and testable research priorities for connecting evaluation to deployment decisions. The framework is an evidence-appraisal proposal whose usefulness and operational benefits require separate validation. An online companion supports inspection and reuse through a searchable bibliography,

Table 1. Adjacent surveys: organizing emphases and the complementary focus of this survey. Symbols represent qualitative author judgments, not rankings of coverage or quality.

Survey and main focus	Emphasis profile					Complementary focus here
	C	B	M	R	D	
Schmidt and Wiegand (2017) [132] <i>Hate-speech features, resources, and NLP methods</i>	●	▲	○	○	▲	Extending content evidence to generated behavior
Fortuna and Nunes (2018) [26] <i>Definitions, datasets, and automatic text detection</i>	●	▲	○	○	▲	Evidence required after policy or observation changes
Garg et al. (2023) [33] <i>Bias sources, targets, and mitigation</i>	●	▲	○	○	▲	How errors affect decisions across configured systems
Wang et al. (2025) [151] <i>Privacy, controllability, authenticity, and compliance</i>	▲	▲	●	▲	○	Support for toxicity claims beyond risk taxonomies
Dan et al. (2026) [15] <i>Multilingual detoxification: data, training, decoding, and alignment</i>	▲	●	○	▲	▲	Action validity under language and policy transfer
This survey <i>Claim validity across content, interaction, visual generation, and components</i>	●	●	●	●	●	Supported claims and required transport tests

● Central ▲ Secondary ○ Non-primary; not necessarily absent.
 C: constructs; B: behavior; M: multimodality; R: reusable components; D: decisions.

an interactive SCOPE worksheet, four worked claim profiles, and reusable evaluation templates.¹ Its source repository provides downloadable records and data-generation scripts.

Section 2 explains the review design; Section 3 defines and applies SCOPE. Sections 4–7 cover content, interaction, visual generation, and components. Sections 8–10 integrate adaptive evaluation, decisions, and the research agenda.

2 Review Design and Evidence Coding

We conducted a structured integrative review covering 183 works in NLP, vision, security, HCI, and model editing. All 183 cited works are included in the descriptive corpus maps. The review does not claim exhaustive coverage.

2.1 Corpus Construction and Eligibility

The corpus cutoff was 23 August 2026. Sources were ACM Digital Library, ACL Anthology, IEEE Xplore, CVF Open Access, PMLR, OpenReview, USENIX, and arXiv. Table 2 records concept families rather than exact historical query strings; citation tracing connected terminology missed by keyword lookup. Eligibility follows the distinction between toxicity endpoints and mechanisms relevant to their measurement or control, rather than treating every safety study as toxicity evidence.

Evidence roles attach to claims. *Direct evidence* defines or measures a retained toxicity endpoint; *tested bridge evidence* compares source and target conditions for that endpoint under Section 3’s criteria; *contextual evidence* explains architectures, attacks, or governance mechanisms without establishing toxicity performance. General safety outcomes are not thereby reclassified. Duplicate versions share one bibliographic record. Figure 2 records 182 peer-reviewed works and one guard-system technical report across the complete bibliography; publication status does not imply a direct toxicity experiment.

¹Website: <https://xukun12138.github.io/toxicity-evidence/>; source repository: <https://github.com/xukun12138/toxicity-evidence>.

Table 2. Search concept families and relevance criteria used in the review.

Family	Search concepts	Evidence sought	Main exclusion
Construct	Toxicity, hate, abuse, harassment, identity attack	Definitions, corpora, annotation, social context	Sentiment without harm policy
Generative behavior	LLM or VLM safety, moderation, guard, refusal	Prompt, response, session, policy transport	Capability without safety outcome
Adaptive access	Red team, jailbreak, suffix, fuzzing, bypass	Threat models, budgets, judge validity	Attack unrelated to governed harm
Visual generation	T2I safety, NSFW, meme, multimodal toxicity	Composition, sampling, visual controls	Generic image quality or watermarking
Components	Embedding, textual inversion, LoRA, backdoor, erasure	Reusable carriers and edited model states	Unrelated personalization
Decisions	Calibration, abstention, review, monitoring, appeal	Evidence-to-action validity and burden	Metrics with no decision relevance

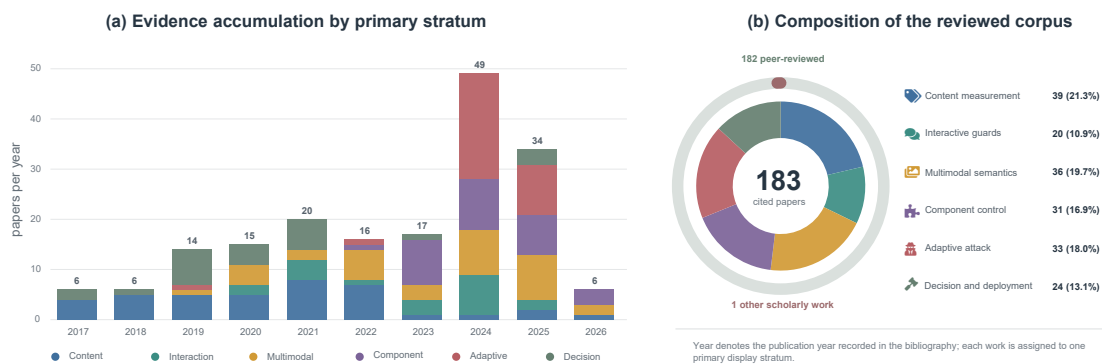


Fig. 2. Publication years and composition of the 183 works included in this review. Panel (a) reports yearly counts under one mutually exclusive display stratum; panel (b) reports 182 peer-reviewed publications and one technical report together with stratum shares. Counts describe the analyzed set, not field prevalence or evidence strength.

2.2 Claim-Level Coding and Review Limitations

Claim-oriented reading records each result's object, mechanism, population, policy, channels, observation/judge, sampling or attack regime, limitation, and proposed action. Endpoints and configurations within a paper require separate interpretation despite its single display stratum. Numerical comparisons retain reported denominators, aggregation, and configurations. Section 3.3 provides four worked profiles. Surveys count as publications, not independent replications of their cited studies: publication counts and claim support serve different purposes.

Support status and evidence quality are separate. Directness to the endpoint, judge validity, precision, coverage, and independent replication inform quality without being collapsed into a score. Components particularly require this distinction: controllability or triggered behavior on named bases does not establish a general toxicity prevalence or release guarantee [29, 80, 89, 163].

Bibliographic details and central quantitative claims were checked against accessible primary reports and publication records where possible. Complete retrieval/exclusion histories, item-level verification logs, independent recoding of all studies, and independently reproduced assignments for Figures 2, 3, and 8 are not provided. Screening/coding agreement

was not estimated; the inspectable corpus does not establish search recall. Publication, accessibility, language, version, and author-familiarity biases constrain coverage. Corpus counts establish neither field prevalence nor evidence strength.

3 The SCOPE Framework: Defining Governance Validity

A sound evaluation can support an unsound deployment inference when its policy, channel, observation, regime, or action changes. SCOPE makes that distinction explicit. Examples include replacing conversation evidence with comment accuracy, refusal appropriateness with refusal detection, or generator robustness with image classification. These are potential extrapolations, not claims attributed to every cited paper.

For a target claim q and evidence set Z , we record a five-coordinate support profile:

$$A(q; Z) = \langle A_S, A_C, A_O, A_P, A_E \rangle, \quad A_d \in \{\text{direct, bridged, unsupported, unreported}\}. \quad (1)$$

Here S is policy; C is channel and carrier; O is the governed unit, available information, and adjudication; P is population, sampling, version, and adversarial regime; and E is action, threshold, and safeguards. *Direct* means that the target claim stays within the evaluated domain, not that the evidence is strong. *Bridged* requires a source–target comparison retaining an identifiable toxicity endpoint, a declared tolerance, and uncertainty analysis. *Unsupported* denotes an identifiable inferential gap or contrary evidence; *unreported* means the record cannot resolve the coordinate. Contextual mechanisms cannot establish either direct or bridged support. Coordinate-wise bridges do not automatically compose: joint changes need joint evidence.

To apply the profile, first state the target claim and source result; then identify changed coordinates, inspect any comparative test, and record the reason for each status. Keep judge validity, statistical precision, population coverage, and independent replication alongside the profile. A *detector* estimates a policy event; a *judge* labels an experimental outcome; a *guard* couples detection to action; a *mitigation* changes behavior. Neither a five-coordinate match nor a low detector score certifies system safety.

3.1 Coordinates of a Governance Claim

A governance claim first requires an identifiable policy event. Hate often involves a protected target; offense need not; harassment depends on directedness and history, while sexual language can be abusive, clinical, or reclaimed. Dataset taxonomies differ [2, 18, 93]; binarization aligns output formats, not necessarily meanings [3].

Let policy π map a realized outcome Y , context k , and target g to $H_\pi(Y, k, g) \in \{0, 1\}$. Arguments are suppressed when the governed unit is clear. When policy distinguishes harm levels, severity is recorded separately as s_π and the ordinal-to-binary mapping is stated. Evaluation and deployment must share π or test their mapping and uncertainty. A policy crosswalk records the source category, target category, inclusion rule, and excluded or indeterminate cases. Hate/harassment/threat categories may enter core synthesis; an NSFW aggregate requires category-level review; object/style backdoor success remains contextual unless its target violates the retained toxicity policy. Unmapped categories are reported separately, never silently pooled.

The next distinction concerns how behavior changes and where harm becomes observable. A *control channel* lets a user, data source, component creator, or integrator alter behavior through prompts, history, images, retrieval, tools, decoding, embeddings, adapters, or fine-tuning. A *risk carrier* holds the evidence in content, an interaction, a multimodal composition, a component, or the deployed system. These roles need not coincide: a benign prompt can yield a toxic response, and an adapter can activate harm only after loading. Figure 3 illustrates this relation. Channel coverage

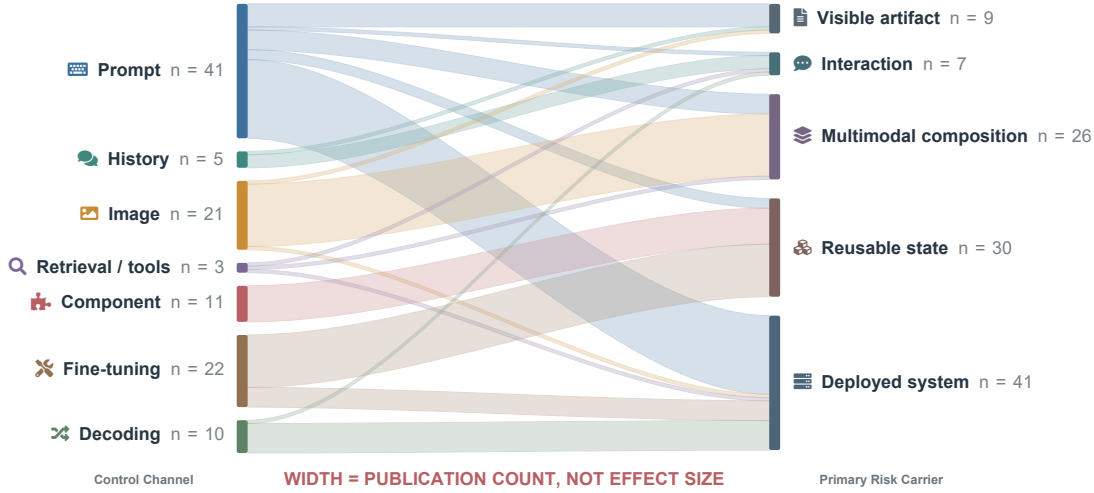


Fig. 3. Control channels and primary risk carriers in the 113-work channel-specific subset of the 183-work bibliography. Each publication contributes one display edge based on its principal manipulation and evidence unit. Reviews, enabling methods, artifact-only studies, and governance analyses are included in the synthesis but excluded from the channel-carrier counts. Ribbon width denotes publication count, not effect size, sample size, incident frequency, attack success, or estimated risk.

means testing named routes under stated conditions. Artifact classification may lack a manipulated channel, whereas a channel experiment may measure its outcome as a visible artifact.

Observation then specifies the governed event, information available to the judge, and unit of adjudication: an artifact, request, response, refusal, trajectory, or executed component. The probability regime determines how observations are sampled and aggregated. For example, judging one image and selecting the worst image from repeated attempts use related observations but estimate different events. Let m denote model and component state, c an access channel, x an input family, η generation settings, a an adversary or sampling procedure, k policy-relevant context, and g the target. With Y generated under these conditions, operational risk is

$$r_{\pi}(m, c, x, \eta, a, k, g) = \Pr(H_{\pi}(Y, k, g) = 1 \mid m, c, x, \eta, a, k, g). \quad (2)$$

Equation 2 defines behavior risk before observing the outcome. After realization y , a detector observes $o = \phi(m, c, x, y, k)$ and estimates the distinct posterior $\Pr(H_{\pi} = 1 \mid o)$. A similarity score need not estimate either quantity. One output cannot characterize seed variation, fixed jailbreak sets do not represent adaptation, and one-base component tests do not identify post-merge behavior [35, 62, 94].

Finally, an enforcement action connects errors to consequences. Detection can inform allowing, warning, refusing, re-generating, reviewing, restricting, or removing, but their severity, reversibility, and safeguards differ. Evidence sufficient to prioritize human review may therefore be insufficient for automatic account restriction. Accuracy and AUC do not resolve this distinction: thresholds require error costs and review capacity, while calibration links probability to frequency, functional tests reveal hidden behavior, and XSTest and multimodal evidence expose the unsafe-compliance/over-refusal trade-off [44, 121, 124, 152]. Section 9 develops the resulting selective decisions.

Table 3. Author-proposed reporting information for falsifiable SCOPE claims and the extrapolations it is intended to expose.

Dimension	Proposed evidence record	Failure class	Unsupported conclusion
Safety policy	Categories, targets, contexts, benign-sensitive cases, annotators, disagreement, crosswalk	Policy substitution	One label set certifies every harm policy
Control channel	Exposed routes, carriers, access, compositions, and per-route tests	Channel omission	Prompt robustness certifies tools, images, or components
Observation unit	Governed event, unit, judge, adjudication, and indeterminate cases	Unit substitution	Response toxicity equals intent, refusal quality, or system safety
Probability regime	Population, sampling, adversary, budget, stopping, version, and uncertainty	Regime extrapolation	One seed or failed finite search establishes durable risk
Enforcement action	Threshold, loss, abstention, capacity, group burden, appeal, monitoring, and expiry	Action escalation	High AUC alone justifies automatic blocking or release

3.2 Support Status and Validity Failures

Comparing source and target coordinates distinguishes missing reporting from an identifiable unsupported extension. The latter exposes five failure classes: policy substitution, channel omission, observation-unit substitution, regime extrapolation, and action escalation. Channel substitution leads to channel omission when evaluation does not test the alternative route through which harmful behavior can occur; Figure 1 illustrates this mechanism. Table 3 gives an author-synthesized reporting checklist, not a validated certification standard. A study need not cover every channel or action, but a positive claim should state its domain and the evidence needed for broader use.

These classes describe *what* becomes unsupported; channel substitution, repeated-access amplification, and judge dependence help explain *how* such gaps recur. They are complementary descriptions, not an exhaustive causal model or a one-to-one mapping. Section 10 develops their implications for an assurance loop requiring empirical validation.

3.3 Worked Claim Profiles and Transport Tests

The following applications audit deliberately extended claims, not the original authors' wording. Profiles are ordered $\langle S, C, O, P, E \rangle$; D, U, and N denote direct, unsupported, and unreported. None is asserted to establish a successful empirical bridge.

C1: Functional accuracy to conversation blocking. HateCheck tests 29 linguistic functions in constructed English cases [124]. Consider the target claim that these results justify automatically blocking English conversations under the same hate policy. Its profile is $\langle D, U, U, U, U \rangle$: policy is held fixed, but conversational history, session adjudication, traffic, and blocking consequences are not tested by the functional suite. A bridge would pair artifact and session observations on an independently adjudicated conversation sample, retaining benign-sensitive controls and threshold-specific errors. Functional diagnosis remains a valid use of the original evidence.

C2: Refusal detection to justified refusal. WildGuard separately labels prompt harmfulness, response harmfulness, and refusal [45]. For a claim that refusal-detection F1 alone establishes justified blocking on those same exchanges, the profile is $\langle D, D, U, D, U \rangle$. Policy, input access, and evaluation population are fixed, but identifying a refusal does

not judge its appropriateness or validate a blocking rule. The missing comparison combines request policy, response content, safe alternatives, and consequences; it is not merely a new threshold on the refusal label.

C3: One output to repeated-access exposure. For the hypothetical claim that an image-level failure rate also estimates harm within 100 attempts, with policy and interface fixed and no action specified, the profile is $\langle D, D, U, U, N \rangle$. Here N records that the claim specifies no action, rather than that evidence for a named action is missing. The identifiable gaps concern the changed event and access regime. Equation 4 supplies an analytic conversion only for fixed independent trials; a measured extension requires prompt-level repeats, a stopping rule, and uncertainty at the prompt or session level.

C4: Static screening to component release. Consider extending Concept QuickLook’s single-file embedding inspection to automatic release across arbitrary compatible compositions while retaining the study’s NSFW categories [163]. The profile is $\langle D, U, U, U, U \rangle$: D preserves the source NSFW endpoint; it does not establish a mapping to targeted social harm. Execution paths, generated outcomes, composition distribution, and release costs remain unsupported by static similarity. Inconsistent metric definitions additionally reduce confidence in the source result, separately from coordinate support. A bridge would test scanner-selected and scanner-cleared files on declared bases, scales, triggers, and seeds with independent image adjudication and benign controls.

Transport tests preserve the source claim while assessing its extension. Prespecify acceptable degradation δ in policy-relevant error or loss, fix both configurations, and use matched cases where possible. Estimate harmful misses and benign-sensitive false positives separately, with uncertainty at the sampling unit accounting for correlated outputs. Bridged status requires a bound supporting the declared tolerance; an inconclusive interval does not establish equivalence. The target must also meet the action’s absolute error or loss requirement: preserving inadequate source performance cannot justify deployment. A bridge covers only the coordinate changes tested.

4 Content Measurement: From Labels to Policy-Conditioned Evidence

Content detection offers reusable evidence about policy labels, representations, and error patterns. Its methods differ in which missing signal they address: domain language, target/context, rare examples, or explicit policy instructions. These differences matter when a detector is incorporated into a generative pipeline.

4.1 The Label Is a Policy Decision, Not a Natural Kind

Datasets operationalize different events. Davidson et al. separate hate from offense; Founta et al. distinguish abusive, hateful, spam, and normal; HatEval restricts targets to immigrants and women; OffensEval hierarchically represents offense, attack type, and target [2, 18, 28, 173]. Abusive-language subtasks also differ by target and explicitness [156]. Collapsing them into toxic/non-toxic changes policy π before training.

Surface form alone cannot resolve these policy distinctions. Latent Hatred adds implicit categories and implied statements; HateXplain separates class, target, and rationale; Toxic Spans localizes evidence [23, 93, 111]. Their supervision exposes different missing information: implied meaning, directedness, or evidential location. Meta-toxic knowledge retrieval supplies relations absent from a prompt, while introducing dependence on retrieval coverage and errors [182]. A plausible rationale can aid diagnosis without proving causal reasoning. Reuse therefore needs to preserve (π, k, g) —policy, context, and target—so that abuse remains distinguishable from profanity, quotation, or counterspeech. Population, language, sampling, and annotation further constrain transfer because resources concentrate in a few platforms and languages [4, 5, 113, 146].

Table 4. Representative content evidence through SCOPE; counts describe resources, not platform prevalence.

Evidence	Policy and observation unit	Source / control channel	Probability regime and scale	Bounded claim and principal limitation
Davidson et al. [18]	Hate, offensive, neither; tweet	Lexicon-sampled Twitter	Enriched English sample	Diagnoses hate/offense confusion; not prevalence and may miss coded hate.
Founta et al. [28]	Abusive, hateful, spam, normal	Iterative Twitter crowdsourcing	~80K, minority enrichment	Scalable construction; enrichment/protocol condition rates.
HateXplain [93]	Class, target, rationale	Twitter and Gab	Human-labeled sample	Joint diagnosis; plausible rationale need not be causal.
HateCheck family [122, 124]	Linguistic function; contrast	Designed benign/hateful cases	HateCheck: 29 functions, 3,728 cases; multilingual: 34 functions, 36,582 cases in 10 languages	Diagnoses represented functions; not prevalence or exhaustive robustness.
DynaHate / Hate-moji [64, 147]	Adversarial word-ing/composition	Human attacks on detectors	DynaHate: 41,255 entries, including ~15K contrastive perturbations; HatemojiCheck: 3,930 cases	Iterative discovery; creators, seed models, and functions bound coverage.
ToxiGen [46]	Text about 13 groups	Machine-generated; targeted human annotation	274,186 statements; 792-item validation test and 8,960 human-labeled training samples	Synthetic stress evidence; full corpus is not uniformly adjudicated.
ToxicChat [77]	Prompt toxicity/jailbreak	Vicuna prompts	demo 10,166; one service/period	Shows social-to-AI shift; not all policies or output harm.

4.2 A Dataset Is a Measurement Instrument

Sampling determines which errors a dataset can reveal. Keywords enrich lexical cues but miss euphemism; source-based sampling can leak identity; random samples contain few rare severe events. Founta et al. enrich minority classes in about 80K tweets, trading population representativeness for more observations of minority labels [28]. Wulczyn et al. label over 100K Wikipedia comments and apply a model to about 63M; this extends coverage through predictions, not additional gold annotation [160]. These strategies address scarcity differently, and neither corpus size nor class proportions alone establish platform prevalence.

Annotation adds another layer. Majority vote can erase substantive disagreement. African American English (AAE) tweets were more likely to be rated as offensive, and providing dialect and racial context partly reduced this disparity. Controlled studies also link annotator beliefs to ratings of anti-Black language, AAE, and vulgarity [129, 131]. Studies of disagreement, annotation paradigms, and configurable juries show that some variation reflects perspective or the choice between population description and policy compliance, not random label noise [42, 100, 123, 144]. Thus label z is not merely noise around one latent truth:

$$z_{ij} \sim p(z \mid x_i, \pi, k_i, g_i, a_j, v). \quad (3)$$

Here a_j denotes annotator perspective and v the protocol. Equation 3 is an explanatory model, not a fitted process. Population-description tasks retain vote distributions; policy-enforcement tasks also require explicit adjudication, including unresolved cases.

Table 5. Detection mechanisms and the information they add. Conditions and costs are synthesis judgments, not a cross-paper performance ranking.

Mechanism / evidence	Useful signal and operating condition	Paired test and principal cost
Supervised/domain-adapted encoding [8]	Learns task labels and domain-specific language	Cross-source holdouts; retraining after domain or policy change
Adversarial/synthetic data [46, 147]	Covers rare or implicit patterns missed by existing models	Independent natural samples; generator/filter dependence
Target, context, and rationale modeling [93, 110]	Resolves directedness and evidence within visible context	Context/target swaps; annotation and context-access burden
Policy-conditioned or retrieval-augmented judging [180, 182]	Makes rules or missing semantic relations available at inference	Held-out policy/knowledge tests; latency and retrieval error
Joint prompt–response labeling [45]	Separates request harm, output harm, and refusal	Joint benign/harmful controls; end-to-end pipeline cost

Table 4 shows why size alone is insufficient: natural, enriched, synthetic, and functional-test samples identify different regimes. Beyond datasheets [34], the record needs policy instructions, sampling queries, platform/date, dialect coverage, enrichment, disagreement, and exposure to removed content.

4.3 From Random Splits to Failure-Oriented Tests

A split of independent, identically distributed (IID) examples estimates performance on a constructed distribution. HateCheck, Multilingual HateCheck, and Hatemoji instead diagnose linguistic functions and symbol composition [64, 122, 124]. Toxic Chinese probes visual, phonetic, and structural obfuscations and benign overcorrection [165]. Identity, dialect, multilingual, and context studies expose failures hidden by pooled F1; concatenating context does not establish relational understanding [17, 105, 108, 110].

The resulting diagnosis should guide the intervention. DynaHate uses human attacks across four rounds to expose detector failures [147]; HateBERT adapts representations to abusive-language domains, whereas ToxiGen expands difficult implicit examples through synthetic generation [8, 46]. These approaches address discovery, register mismatch, and coverage, respectively, so gains cannot be interpreted without a corresponding test of the targeted weakness. Table 5 connects mechanisms to these tests. Natural holdouts estimate error on deployment-like cases, contrasts test invariances, and adaptive discovery probes reachable failures. Multilingual mitigation spans training, editing/decoding, alignment, and data construction [15]; translation alone cannot establish cross-cultural policy equivalence.

Separate translated, meaning-matched pairs from independent native-language cases: the former test decision invariance under an agreed policy; the latter expose local expressions and targets missed by translation. Distinguish translation failures from substantive policy disagreement before assigning detector error. Match interventions to the diagnosis: adapt representations for linguistic mismatch, add missing context, or revise policies to address conflicting judgments. Repeat diagnostic contrasts and independent native-language tests to detect new errors; improvement on repaired examples alone does not establish transfer.

4.4 Metrics Must Match the Enforcement Action

Diagnostic gains support an action only when evaluated at its operating threshold. Identity-term tests and subgroup/background AUCs expose different false-positive and false-negative mechanisms [6, 22], but represented identities bound that diagnosis and ranking performance supplies no normative threshold. Threshold selection also requires

error costs and review capacity, followed by precision and recall with prevalence and uncertainty. Calibration should be checked on adequately sampled policy/language slices rather than presumed from a global fit [44]; sparse slices warrant uncertainty or review. Thus ranking, calibration, and threshold-specific error answer complementary decision questions.

Optimization can change represented language. Detoxification improved automatic scores on RealToxicityPrompts while reducing coverage of marginalized groups/dialects, and human raters disagreed with many high scores after strong intervention [157]. Structured social-bias frames show why target and implied social meaning can be part of the observation rather than an optional explanation [130]. Evaluation therefore needs policy recall, benign-sensitive false positives, group/language error, calibration, and utility, not a scalar “toxicity reduction.” Community-moderator evidence likewise shows that generic toxicity labels cover only part of an enforcement workflow’s rules and support needs [7]. Content recognition can therefore support triage under matched policy, observations, distribution, and threshold; moving to conversation-level action requires the additional evidence illustrated by C1.

5 Interactive Behavior: From Output Scores to Trajectory-Level Safety

For assistants, request intent, sampled completion, refusal, and cumulative dialogue harm define different targets. Input filtering, response filtering, and session escalation therefore require different observations; a binary label alone does not identify the supported decision.

5.1 Generation Creates a Conditional Event Distribution

RealToxicityPrompts scores 100K natural English prompts and repeatedly samples several language models, showing reachable toxic continuations even from relatively innocuous prompts [35]. HONEST instead measures hurtful sentence completions through identity templates [103]. Natural prompts expose behavior under observed language; templates isolate identity-conditioned contrasts. Their reported rates depend on the models, decoding settings, number of attempts, identity coverage, and outcome judges, which must accompany comparisons.

The distinction matters operationally. If a prompt x produces an unsafe completion with per-sample probability $r_\pi(x)$ under fixed settings, then n independent attempts expose at least one unsafe output with probability

$$R_\pi^{(n)}(x) = 1 - (1 - r_\pi(x))^n. \quad (4)$$

Figure 4 illustrates Equation 4, not observed attack performance. Report unique prompts, samples per prompt, total outputs, and sessions separately. Averaging prompt-level rates before applying the nonlinear formula generally gives a different result from averaging prompt-specific repeated-access risks. Correlated or adaptive attempts require their own sampling analysis.

Under the same Bernoulli model, a fixed budget of N independent trials with correct adjudication yields a useful non-discovery bound. Since the probability of observing no violations is $(1 - r)^N$, zero observed violations gives the exact one-sided $(1 - \alpha)$ confidence upper bound $1 - \alpha^{1/N}$ for the per-trial rate, not a rate of zero. This calculation concerns the specified prompt and configuration; it does not bound unseen prompts or an unrestricted attack class. Adaptive stopping and judge errors require additional analysis rather than substitution into the same formula.

PPLM, GeDi, and DExperts steer output distributions at inference time [16, 65, 78]. When a related classifier both guides and scores decoding, gains may reflect judge agreement. Independent audits of topic, dialect, diversity, and meaning must check whether benign content is suppressed [157].

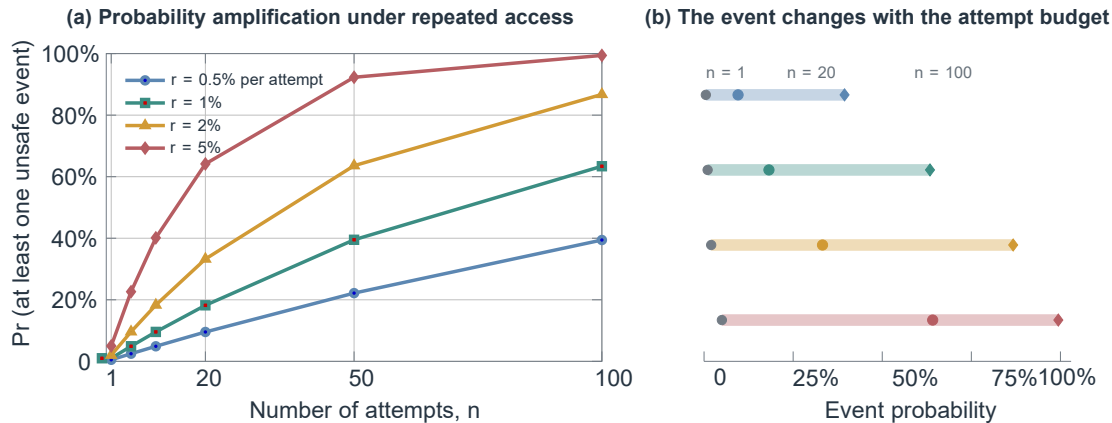


Fig. 4. Probability amplification under repeated access. Values are derived from Equation 4 for illustrative per-attempt rates under an independence approximation; they are not empirical safety estimates. Panel (b) contrasts the event at one, 20, and 100 attempts.

5.2 Separate Labels for Harm and Refusal

Three labels are useful: request harmfulness, response harmfulness, and whether a refusal occurred. *Refusal appropriateness* is a further judgment requiring the request’s policy status, the information released, and permitted safe assistance. A refusal can leak harmful details; a safe response need not refuse. C2 therefore separates label prediction from justified intervention.

Llama Guard documents policy templates and input/output classification as a technical report [57]. WildGuardMix contains about 87K training examples and 5,299 human-annotated test items across 13 risk categories; its tasks are prompt harmfulness, response harmfulness, and refusal detection [45]. The test count combines task-specific items; it should not be interpreted as 5,299 independent conversations. In WildGuard’s Table 5 ablation, removing synthetic adversarial data reduces adversarial-prompt F1 from 85.5 to 77.1 within the same protocol. This supports attack-matched data coverage, not immunity to new attacks or policy changes.

SALAD-Bench structures harm categories, ShieldLM conditions judgments on policy, and SORRY-Bench probes refusal [74, 162, 180]. Their taxonomies require crosswalks. Input, output, and tool-call guards differ in available evidence and intervention delay. Table 6 organizes regimes needing end-to-end residual-harm, benign-utility, and latency tests. Paired comparisons should hold exchanges fixed while varying policy, then hold policy fixed while varying responses to test harmfulness and refusal. Refusal-label agreement cannot replace these comparisons; guard placement determines what evidence is available at intervention.

5.3 Alignment Changes Behavior but Does Not Close the Channel

Alignment changes output distributions rather than classifying individual inputs or outputs. In InstructGPT’s human-comparison protocol, the 1.3B aligned model was preferred to the 175B GPT-3 baseline and generated less toxic text on tested prompts [106]. This establishes a within-protocol preference, not a general capability ranking. BeaverTails separates helpfulness and harmlessness through 333,963 QA pairs with safety meta-labels and 361,903 expert comparison pairs [59]; Safe RLHF makes the safety constraint explicit instead of folding it into one preference objective [14]. Explicit objectives clarify training, but neither study establishes their policy adequacy for a particular deployment.

Table 6. Interactive-safety evidence through SCOPE; all counts and claims are configuration-bound.

Evidence regime	Policy / observation unit	Channels and probability regime	Judge and action	Supports; does not certify
RealToxicity-Prompts [35]	Sampled continuation	100K natural prompts; repeated samples	Automated score; compare/intervene	Reachability under protocol; not model-wide rate or conversation safety.
WildGuard [45]	Prompt harmfulness, response harmfulness, refusal	About 87K train; 5,299 human test; 13 risk categories	Guard; allow, block, score	Represented policy performance; not attack or policy-transfer immunity.
HarmBench / JailbreakBench [9, 94]	Target behavior elicited	Standardized behaviors, attacks, targets, budgets	Benchmark judge; compare attacks	Reproducibility in threat model; not unseen-attack bound or prevalence.
StrongREJECT [138]	Relevant harmful information	313 forbidden prompts; attack outputs	Human-aligned effectiveness judge	Detects inflated ASR; not complete policy or perfect oracle.
XSTest / OR-Bench [13, 121]	Refusal appropriateness	Benign-sensitive/toxic contrasts; OR-Bench: 80K benign, ~1K hard subset, 600 toxic controls	Answer, refuse, review	Joint over-refusal/compliance; not universal benignity.
Crescendo [127]	Dialogue-level reachability	Adaptive escalation; up to 10 rounds	Session judge/action	Multi-turn composition; not absolute attack probability.

Preference coverage and persistence need separate tests. Multilingual jailbreak evidence motivates language-stratified evaluation beyond English alignment [20]. Fine-tuning GPT-3.5 Turbo on 10 adversarially designed examples substantially compromised evaluated safety; benign tuning also caused degradation [115]. Alignment evidence therefore attaches to tested languages and model states; subsequent tuning requires reassessment.

5.4 Adversarial Trajectories and Two-Sided Evaluation

Red-team evidence depends jointly on attack generation, search protocol, and success judgment. Human, model-generated, gradient, fuzzing, and search attacks explore different regions; model-generated red teaming scales discovery while creating an attacker–target–judge dependence that needs auditing [112]. HarmBench compares 18 red-team methods across 33 target/defense configurations, and JailbreakBench standardizes 100 behaviors and their attack/scoring artifacts [9, 94]. Protocol comparability still depends on meaningful success judgments. StrongREJECT’s 313 prompts require specific forbidden information and expose ASR inflated by empty or irrelevant replies [138]. Refusal bypass, policy violation, harmful-information quality, and downstream success should therefore remain separate outcomes.

History changes attack surfaces and intervention timing. Crescendo escalates apparently benign turns using prior replies, with automated tests capped at 10 rounds [127]. LLaVASHield covers 4,484 multimodal multi-turn samples across eight primary safety dimensions and 60 subdimensions [55]. Session state and cumulative cross-modal meaning matter, but retrospective labels do not establish deployable warnings. Prospective tests must use only available dialogue prefixes and measure intervention before disclosure; later refusal cannot retract information. Paired benign trajectories test premature blocking. Report detection timing, cumulative released harm, and final-session accuracy under fixed reset and stopping rules.

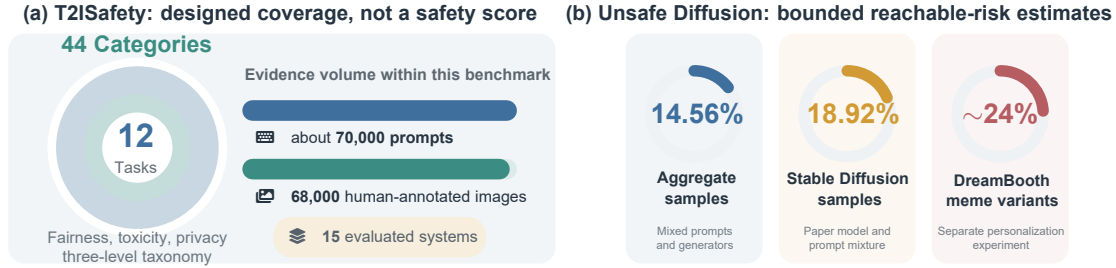


Fig. 5. Two kinds of quantitative evidence in visual safety. Panel (a) summarizes T2ISafety’s 12 tasks, 44 categories, approximately 70,000 collected prompts, 68,000 annotated images, and 15 systems listed in its Table 4 [75]. The evaluated subset is 2,669 prompts, not the full resource pool. Panel (b) reports Unsafe Diffusion’s 14.56% across 68,832 outputs from four generators on the same 17,208-prompt mixture, 18.92% among the 17,208 Stable Diffusion outputs, and 24% among 100 DreamBooth variants in a separate experiment [116]. The first two share the prompt mixture and policy but differ in generator aggregation; the third has a different object and denominator. These conditional sample frequencies should be kept separate and do not estimate deployment prevalence.

Benign controls are equally necessary: indiscriminate refusal blocks legitimate identity, health, abuse, or security discussion. XSTest pairs sensitive benign prompts with unsafe controls; OR-Bench adds 80K benign prompts, a roughly 1K hard subset, and 600 toxic controls across 32 LLMs [13, 121]. Toxic controls prevent “answer everything” from appearing optimal. WildTeaming mines 5.7K unique clusters of in-the-wild jailbreak tactics and constructs WildJailbreak, a synthetic corpus of 262K prompt–response pairs spanning vanilla and adversarial cases with harmful and benign contrasts [61]. Its paired coverage remains bounded by source prompts, clustering, and generators, requiring naturalistic transfer tests.

Joint evaluation should report unsafe compliance, inappropriate refusal, and uncertainty by policy, language, and interaction length. Improved harmful-request recall alone cannot establish a better workflow when sensitive benign requests are indiscriminately blocked. The unit of assurance is thus the configured model–guard–history pipeline, including tools, sampling, and thresholds; Section 8 compares its attack regimes.

6 Compositional Semantics, Generation Risk, and Visual Safety

Visual safety spans image–text artifact classification, VLM response judgment, and harmful T2I generation. A meme requires relational observation; a generator requires sampling under a configuration. Figure 5 separates coverage resources from conditional sample frequencies.

6.1 Composition Defines the Observation

Let I denote an image, T its visible or associated text, k its context, and g the represented or addressed target. The moderation object is

$$H_{\pi} = h_{\pi}(I, T, k, g). \quad (5)$$

Equation 5 is policy-conditioned: the same artifact may receive different labels under different rules or contexts. Sexual content, violence, and general unsafe labels enter toxicity claims only through the crosswalk in Section 3.

Hateful Memes replaces images or text to construct benign confounders and suppress unimodal shortcuts [63]. Its roughly ten thousand English memes and binary policy differ from unrestricted traffic. MMHS150K offers larger, naturalistic Twitter data, but its tested multimodal systems did not outperform a text-only alternative [40]. SIUO’s 269 cases (167 human-crafted, 102 AI-assisted) span nine safety domains where modalities that appear safe separately can

jointly elicit unsafe VLM output [150]. Controlled pairs test relations, platform data test visual evidence amid noise, and SIUO tests cross-modal alignment. These designs cannot generally rank text-only against multimodal models. Comparisons must establish whether the visual relation is necessary for the retained label, then assess its use under the intended input distribution.

VLM safety benchmarks extend this distinction from artifact classification to generated behavior. MM-SafetyBench evaluates 5,040 image-text pairs over 13 scenarios and 12 models, while RTVLM spans 12 red-teaming subtasks and ten open-source VLMs [76, 84]. Their policies, transformations, systems, and judges differ. Together they show that visual instruction creates an additional control channel whose relational failures require task-specific measurement.

Annotation targets constrain useful representations. Memotion models pragmatic signals, MAMI misogyny subtypes, and MOMENTA/BHM harmfulness with targets [25, 51, 114, 136]. ViLBERT, CLIP, BLIP, and LLaVA provide enabling representations, not toxicity validation [73, 79, 87, 118]. Hate-CLIPper learns cross-modal feature interactions, while retrieval-guided contrastive learning supplies relevant comparisons for named meme benchmarks [68, 96]. These approaches change feature interaction and comparative evidence, respectively; their benefits depend on the ontology and feature, retrieval, and annotation costs.

Relational tests should swap one modality, paraphrase while preserving the target/implication, and vary explicit policy while holding the artifact fixed. Slice results by language, template, target, and OCR quality. Context visible to annotators must also be visible to the judged system; otherwise disagreement can reflect missing information rather than model failure.

6.2 Generation Converts a Label into a Risk Probability

Diffusion foundations explain how a condition reaches an image [48, 102, 120], and LAION-5B explains part of the inherited web-data exposure [134]; neither alone measures toxicity. For generator version v , condition q , sampling configuration η , context k , and target g , the safety object is

$$p_{\pi}(q, \eta, v, k, g) = \Pr_{I \sim G_v(\cdot | q, \eta)} [h_{\pi}(I, q, k, g) = 1]. \quad (6)$$

Equation 6 shows that changing a negative prompt, guidance scale, checkpoint, checker, image condition, or adapter changes the claimed system. One curated output estimates neither p_{π} nor exposure under repeated attempts.

Unsafe Diffusion reports 14.56% across 68,832 outputs, with 17,208 per generator, and 18.92% for Stable Diffusion [116]. Its 24% DreamBooth frequency uses a separate 100-variant experiment; agreement across all 300 personalization variants was Fleiss' $\kappa = .85$. The five-category unsafe policy, including a political category, is not our core toxicity definition. T2ISafety's resource counts likewise differ from evaluation denominators: its 2,669 evaluation prompts include 1,787 toxicity, 236 fairness, and 646 privacy prompts, while Table 4 lists 15 systems [75]. Whole-benchmark averages must not be relabeled toxicity rates.

Repeated generation requires prompt/seed counts, failures, stopping, and uncertainty. Traffic-weighted means need declared traffic weights; macro averages expose category differences; a bounded maximum describes a search, not prevalence. The independence conditions of Equation 4 do not cover adaptive prompts.

UnsafeBench separates 5,060 LAION-derived and 5,086 Lexica-generated images across 11 categories; VSCBench separates discrimination from confidence reliability [37, 117]. These results motivate source-specific judge audits. Sampling intervals and judge error are different uncertainties: correlated outputs require prompt-level or sequential analysis, whereas annotation error requires independent adjudication.

Table 7. T2I control surfaces, reported failure modes, and proposed paired evaluations.

Control point	Representative evidence	Characteristic failure	Proposed paired evaluation
Prompt / condition	Latent Guard [82]	Obfuscation, paraphrase, learned tokens, or harmful image-text composition	Harm recall and benign-sensitive false positives across unseen concepts, languages, and attacks
Sampling trajectory	SLD; SAFREE; Detect-and-Guide [71, 133, 170]	Unsafe seeds, weakened fidelity, or out-of-subspace harm	Multi-seed residual risk, fidelity, quality, and category coverage
Model weights	ESD; MACE [30, 88]	Prompt recovery [143, 178] or collateral damage	Direct/adversarial prompts and adjacent benign concepts
Output image	UnsafeBench judges [117]	Stylization, source shift, or implicit contextual harm	Independent audit, per-category calibration, and cross-generator transfer
Composed system	Full-stack ablation (proposed)	Correlated errors, ordering effects, shared policy gaps	End-to-end risk and utility with component ablations and fingerprints
Adaptive interface	SneakyPrompt; MMA-Diffusion [166, 168]	Search across text, image, editing, and checker channels	Query-bounded attacks, attempts-to-success, failure logs, and current version

6.3 Control Stacks and Adaptive Evaluation

Controls act on different information and at different stages. Latent Guard classifies the condition; Safe Latent Diffusion, SAFREE, and Detect-and-Guide alter denoising; ESD and MACE edit weights; output filters judge pixels [30, 71, 82, 88, 133, 170]. GuardT2I, VLGuard, ECSO, and AlignGuard respectively reason over embeddings, tune multimodal behavior, convert visual evidence to text, or add alignment adapters [43, 81, 167, 183]. Condition screening can reject requests before generation; later controls inspect realized content, making complementarity depend on surviving failures. Shared data or representations can correlate errors, so multiplying layer scores cannot establish a guarantee. Table 7 connects control points to residual-harm and benign-utility tests within one stack.

The intervention point also determines the relevant attack. SLD controls inference, ESD/MACE suppress concepts in model state, and recovery attacks probe broader claims through another access route [30, 88, 133, 143, 178]. SneakyPrompt uses repeated black-box queries to bypass evaluated DALL-E 2 and Stable Diffusion filters [168]; MMA-Diffusion crosses text and image conditions [166]; Ring-A-Bell and To Generate or Not recover erased targets under declared attacks [143, 178]. Comparison requires matching access, budgets, policy, utility set, judge, and interface.

UnsafeBench motivates source/generator strata and T2ISafety category-specific reporting [75, 117]. Within each stratum, evaluation should separate prompt, image-condition, editing, and output-checker access, then run common cases through the unguarded baseline, individual controls, and complete stack with fixed budgets and independent adjudication. Ordering and ablations reveal whether an added filter supplies complementary coverage or merely adds cost while preserving correlated misses.

For request-level harm rates, the denominator should include all attempted requests, including those rejected by filters or terminated after exhausting regeneration budgets. Report harmful outputs delivered per attempted request, the harmful fraction among delivered outputs, and refusal or regeneration frequency separately. A later checker sees cases selected by earlier controls, so its conditional error cannot be inferred from its standalone benchmark rate. Rejection, exhausted regeneration budgets, and technical failure should remain distinguishable outcomes. This accounting prevents lower output volume from being mistaken for improved discrimination and connects residual exposure to the cost of denying benign service.

7 Hidden Components as Toxicity Control Channels

Embeddings, LoRAs, adapters, checkpoints, and erased states can change behavior after loading. Here *hidden* describes policy effects not fully revealed by ordinary prompt-output observations, not physically inaccessible weights. Because activation depends on the base, tokenizer, loader, strength, triggers, co-components, and sampling, the evidentiary gap

lies between inspecting a portable artifact and observing its execution. Privacy, ownership, and provenance enter only when relevant to that moderation decision.

7.1 Portable Semantics Change the Moderation Unit

Textual Inversion stores a concept in token embeddings [29]; DreamBooth fine-tunes weights from a few images [125]; Custom Diffusion updates cross-attention parameters and composes concepts [70]; and LoRA stores low-rank deltas [52]. These are controllability results, not toxicity evaluations, but they show that semantic capability can circulate separately from a base model and remain dormant until loaded or triggered.

The observation unit is therefore a resolved composition

$$y = G(m \oplus_\lambda u, p, \eta), \quad (7)$$

where m is the base state, u the component, $\oplus_\lambda$ the loading operation and strength, p the interaction, and η the generation settings. Its conditional policy risk is

$$r_\pi(u \mid m, \lambda, \mathcal{P}, \mathcal{E}) = \Pr_{p \sim \mathcal{P}, \eta \sim \mathcal{E}} [H_\pi(G(m \oplus_\lambda u, p, \eta)) = 1]. \quad (8)$$

The loading operator $\oplus_\lambda$ abstracts format-specific execution, not literal addition for every component. A scanner observes $\psi(u)$; Equation 8 instead averages over executed compositions, with sampling randomness included in η . Embedding geometry, LoRA layers/rank, and checkpoint differences are different inspection objects. Base compatibility, co-adapter order, and later tuning prevent static similarity from substituting for behavior.

7.2 What Static Concept Inspection Establishes

Concept QuickLook illustrates the gap between static observation and behavioral evidence through metadata-based Concept Matching and metadata-free Fuzzy Detection [163]. Several survey authors also authored that work. This configuration-bound feasibility study illustrates C4's screening-release gap, not registry defenses generally.

The study covers SD 1.5/2.0 embeddings and 800 reviewed files, 500 downloaded and 300 customized, with 15% marked NSFW. Two annotators review examples and a third resolves disagreements. This sample establishes neither ecosystem prevalence nor per-split class balance. Its reported Accuracy/FPR/FNR values are not jointly reconcilable under conventional binary definitions; Figure 6 therefore reproduces separate reports rather than a recoverable operating point.

QuickLook reports 1.5 seconds and 0.8 GB including initialization/inference, versus 15 seconds for generation-based Matching. Fuzzy generation time is separately reported as 30 seconds and 32.0 ± 2 seconds; generation uses 5.3 GB. These hardware-bound resource comparisons do not calibrate the normalized similarity R_c as $\Pr(H_\pi = 1 \mid o)$: it remains a priority index, and neither its review bands nor the incompatible aggregate metrics support predictive-value or workload calculations.

The resource contrast motivates a *hypothesis*: static triage could reduce costly generation. Testing it requires reconciled metric definitions, representative prevalence, behavioral audits of flagged and cleared files, and total workflow cost. Arbitrary LoRAs, checkpoints, tokenizers, multi-concept bundles, and adversarial files also require tests under their intended execution conditions. The case identifies the missing link between static observation and release authority.



Fig. 6. Separately reported Concept QuickLook metrics and resources [163]. The metric panels reproduce source-table values. Under conventional binary definitions, Accuracy, FPR, and FNR are not jointly reconcilable in any mode or split, precluding derived prevalence, predictive value, and review workload. The time panel reports QuickLook at 1.5 seconds, generation-based Matching at 15 seconds, and Fuzzy at 30 seconds and 32.0 $\pm$ 2 seconds separately. The two Fuzzy timing reports, 30 seconds and 32.0 $\pm$ 2 seconds, are displayed as separate source-reported values. Timings and memory are hardware-specific.

7.3 Dormant Activation and Composition

Themis tests textual-inversion controls against triggers, synonyms, and combinations [159]. LoRATK transfers backdoors through LLM LoRAs, while MasqLoRA hides T2I backdoors behind benign functionality [80, 89]. These studies expose conditional activation across different component formats; their backdoor endpoints become direct toxicity evidence only when the target satisfies the retained harm policy. BadDiffusion, TrojDiff, and few-shot personalization attacks similarly preserve ordinary behavior while changing triggered outputs [11, 12, 56]. Together they motivate clean/triggered comparisons, not registry prevalence estimates from attack success.

The evaluation unit is an activation envelope

$$\mathcal{E}(u) = \mathcal{M} \times \mathcal{A} \times \mathcal{P} \times \mathcal{S} \times \mathcal{C}, \quad (9)$$

covering bases, strengths, prompt and trigger families, stochastic settings, and co-components. Compatibility constrains this product: not every tuple executes. Tests sample an admissible envelope rather than exhaust it; non-triggered and neighboring benign concepts are essential controls. A changed base, tokenizer, loader, order, or policy requires retesting or a narrower claim.

In MasqLoRA's Scenario #1 on SD 1.5, increasing the number of stacked malicious adapters from one to four reduces object-backdoor ASR from 99.8% to 91.6% and the benign-function CLIP score from 31.22 to 27.3 [89]. This experiment measures the persistence of an object backdoor under adapter composition; its ASR does not measure toxicity. The result supports composition-aware testing: behavior can persist in the loaded stack despite benign metadata. Log module order/strength, successful and failed triggers, bases, seeds, and utility.

Attribute component effects by comparing the same base alone, with the component, and, where meaningful, with a benign task-matched component under matched prompts and sampling. Report absolute harm and differences: small

increases can coexist with unsafe bases. This localizes behavior to a tested composition without defining context-free file maliciousness.

7.4 Stateful Mitigation and Component Release

Erasure methods differ in how they redirect a learned concept and preserve surrounding behavior. ESD uses negative guidance, concept ablation an anchor, Selective Amnesia continual learning, and Unified Concept Editing closed-form projections; MACE, All But One, and Receler emphasize scale, preservation, or locality [30, 31, 47, 50, 54, 69, 88]. Comparison must retain the target, replacement or anchor, and neighboring benign concepts. Object/style suppression remains a mechanism result unless the target maps to toxicity.

Direct suppression and benign locality must be evaluated alongside recovery resistance and persistence. Ring-A-Bell and UnlearnDiffAtk recover targets under specified prompt/access conditions [143, 178]. Receler addresses prompt robustness, but an owner with weight access can remove its eraser [54]. Prompt recovery tests the retained state through its interface; eraser removal changes that state. A single robustness score obscures the relevant release restriction. The distinct requirements are summarized by

$$\mathbf{v}_{\text{erase}} = (S_z, U_{\neg z}, R_{\text{prompt}}, R_{\text{component}}, R_{\text{relearn}}), \quad (10)$$

In Equation 10, S_z is suppression, $U_{\neg z}$ preserved utility, and the R terms cover prompt, component, and re-learning attacks. Papers populate different coordinates, leaving one success rate insufficient for ranking them.

Newer methods address saliency, adversarial training, multi-target interference, and local preservation [24, 72, 141, 176, 177]; benchmarks test latent recovery, adaptation, meta-unlearning, and non-text conditioning [32, 38, 99, 126, 158]. Prompt resistance leaves component changes and re-learning untested; utility preservation does not establish cross-channel suppression. Match target/anchor ontology, preservation set, model, judge, access, and budget, leaving untested coordinates unknown. Table 8 connects behavioral, integrity, and operational evidence to release questions. Prompt recovery and eraser removal assume different access when providers retain weights; removal enters the release claim when recipients can modify weights. Admissible use therefore depends jointly on tested persistence and enforceable access, not suppression alone.

A proposed release workflow combines integrity/static checks, sandboxed compositions, and allow, restrict, quarantine, or review decisions. Figure 6 motivates its resource hypothesis, while end-to-end validity still requires sampled outcome audits of cleared and escalated files to detect missed activation. Net savings must account for audit costs and the consequences of missed compositional failures.

Anti-DreamBooth perturbs source images against unauthorized personalization [145]. Conditional on preprocessing and training, this is contextual data-intervention evidence, not released-component toxicity testing. Release manifests should bind artifact identity, compatible execution settings, scanner/threshold, tested prompts, triggers/seeds, outcomes, action, and expiry. Erased states also require targets, anchors, edited modules, and recovery tests. Binding these records to executed states specifies retesting conditions.

8 Adaptive Evasion and Detection–Mitigation Coupling

Across modalities, adaptation couples detection to subsequent inputs through attacker feedback. Its evidence depends on access, optimization unit, budget, stopping, and adjudication. Counterexample discovery, conditional success estimation, and bounded robustness require different accounts of search coverage.

Table 8. Proposed complementary evidence layers for component-level toxicity assurance.

Evidence layer	Test object and procedure	Bounded claim supported	Residual failure	Appropriate action
Identity and integrity	Hash, signature, parser, tensor schema, claimed base and manifest	Artifact is well formed and matches a version	Semantic payload and triggers unknown	Reject malformed files; retain provenance
Static semantics	Embedding class, weight-diff features, prototypes, meta-data consistency [163]	Similarity to known concepts under the scanner	Uncalibrated, OOD, obfuscated, or base-specific score	Prioritize further tests; no release-safety inference
Reference behavior	Declared bases; direct, indirect, trigger, and benign prompts over seeds	No tested violation within stated composition budget	Unseen triggers, bases, merges, or search	Conditional, fingerprinted attestation
Adaptive assurance	Prompt attacks, scale/order sweeps, co-adapters, eraser removal, re-tuning [143, 178]	Resistance to named attacks and budgets with utility	Future attacks and drift	Restrict access/ranges; publish budget and expiry
Operational evidence	Incidents, appeals, subgroup effects, base/policy changes, disagreement	Performance under observed deployment	Rare harms and strategic adaptation	Re-test, quarantine, revoke, or revise

8.1 Threat Models and the Scope of Attack Evidence

A bypass can expose a failure *within* the declared threat model or exploit conditions outside the tested domain. Representation, interaction, composition, and search attacks change form, distribute intent, add channels, or optimize repeated attempts. Whether these changes invalidate an earlier claim depends on its declared domain, not simply on attack success. Multimodal-jailbreak research distinguishes visual encoding, composition, interaction, and access [149]; these categories locate the mechanism being tested but do not supply a common success-rate denominator.

Optimized triggers and model-assisted methods such as PAIR, AutoDAN, GPTFuzzer, and Tree-of-Attacks search different white- and black-box spaces [10, 83, 95, 112, 148, 171]. Cipher, ASCII-art, persuasion, and defense-aware prompting alter representation or search strategy [19, 60, 172, 174]; HarmBench supplies a common protocol within a finite attack family [94]. An ASR advantage may reflect better search or greater opportunity; isolating strategy requires matching access, targets, budgets, and success criteria.

Visual filters and erased concepts face analogous transformations [143, 168, 178]. JailbreakDiffBench standardizes finite configurations of prompt and image checkers, generators, attacks, judges, and protocols without establishing repeated-seed performance [62]; sample counts and seeds remain necessary. FigStep embeds instructions in images; Multi-Modal Linkage splits intent across modalities; distraction/shuffle attacks target safety recognition; VisCRA targets reasoning traces [41, 137, 155, 169, 181]. These test judgment under transformed or distributed policy information in visual text, layout, cross-modal links, or reasoning traces. Mechanism comparison does not make unlike ASRs commensurate.

If $\mathcal{A}(B)$ is the admissible set of attacks under budget B , an operational robustness target can be written as

$$r_{\pi}^{\text{adv}}(m, c, x, \eta, k, g; B) = \sup_{a \in \mathcal{A}(B)} r_{\pi}(m, c, x, \eta, a, k, g). \quad (11)$$

Equation 11 is an ideal supremum, not an empirical maximum. A confirmed success establishes reachability, not a rate of one. Finite failures can bound risk for a specified sampling procedure under statistical assumptions, but cannot

Table 9. Threat regimes and the scope of inference supported by each.

Regime	Actor and access	Supports	Does not support
Static benchmark	Fixed labeled examples	Error on represented policy/distribution	Paraphrase, interaction, shift, attack
Functional suite	One designed capability	Specified invariance/contrast [124]	Prevalence or exhaustive coverage
Human red team	People probe an interface	Observed misuse and policy gaps for stated recruitment and interface	Worst-case risk or user prevalence absent budgets and sampling
Automated black box	Output/judge-guided search	Observed conditional success within a stated query budget and stopping rule [10, 171]	Deployment prevalence; white-box or future-attack robustness
White box	Internal access	Vulnerability under that access [148]	Ordinary-user likelihood
Composition test	Modules combined after release	Interaction/supply-chain failure [80, 89]	Every possible composition

upper-bound an unrestricted future attack class. Table 9 separates the complementary regimes whose scores become comparable only when policy, targets, budget, model version, and judge are held constant.

AutoPrompt and PLA optimize black-box T2I prompts against different exposed filters [86, 90]. Tensor Trust, InjecAgent, and formal prompt-injection evaluation provide *contextual mechanism evidence* about retrieval and tool-action channels, not tested toxicity bridges [85, 142, 175]. A bridge would require a retained toxicity endpoint and a source–target comparison meeting Section 3’s criteria. Report query counts, seeds, failures, stopping, and judge errors separately.

8.2 Coupled Evaluation and Evidence Maintenance

Alignment, refusal geometry, response-derived evidence, and safe decoding intervene at different points [16, 53, 78, 106, 154, 164]. Evaluation must therefore distinguish five roles—data generator, attacker, reward model, guard, and final judge—and disclose shared models and data. Different prompts do not make a shared model statistically independent; optimization may conceal its systematic blind spots.

A proposed judge audit freezes scoring before comparative results are inspected, then independently adjudicates a held-out sample with system identities concealed. Sample both predicted-safe and predicted-unsafe cases, beyond judge-selected disagreements, and retain policy-relevant indeterminate cases. Oversampling severe or ambiguous outcomes aids diagnosis, but aggregate errors require sampling weights. Report sensitivity across independently checked judges when evaluator choice changes an intervention’s apparent advantage.

Immune changes multimodal behavior at inference time; GuardTrace-VL inspects traces linking questions, intermediate reasoning, and answers [39, 161]. Additional observation can help diagnosis while increasing latency, exposure, and attack surface. Erasure similarly requires suppression, recovery, and benign-spillover evidence [30, 143, 178].

These dependencies also determine when evidence expires. A model, policy, interface, or attack change should trigger regression tests with harmful and benign controls, tied to configuration fingerprints, test ownership, and explicit conditions invalidating reuse. Static screens prioritize known signals while red teams search for reachable failures. Maintaining both updates bounded evidence without establishing permanent robustness.

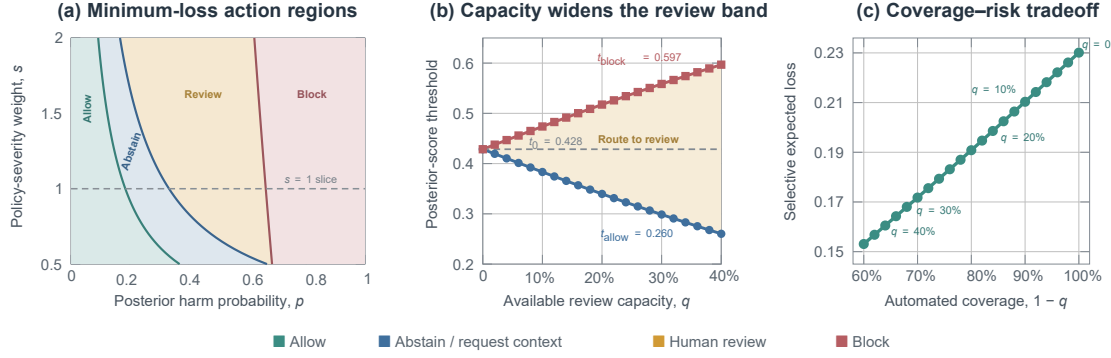


Fig. 7. Decision consequences under two illustrative constructions. Panel (a) minimizes four stated losses: with $p = \Pr(H_\pi = 1 \mid o)$ and normalized severity $s := s_\pi$, $R_{\text{allow}} = sp$, $R_{\text{abstain}} = .12 + .35sp$, $R_{\text{review}} = .22 + .05sp$, and $R_{\text{block}} = .75(1 - p)$. Here abstention withholds an immediate automated decision without assigning a reviewer, whereas review enters a capacity-constrained adjudication queue. Panels (b)–(c) use a separate selective-review construction: at $s = 1$ and $p \sim \text{Beta}(2, 6)$, cases closest to the allow–block indifference point are reviewed until capacity q is filled. They report unreviewed coverage and $\mathbb{E}[\min(R_{\text{allow}}, R_{\text{block}}) \mid \text{automated}]$; they do not optimize the abstain and review losses from panel (a), and the conditional loss excludes reviewer cost and error. All values are synthetic and do not prescribe deployment thresholds.

9 Decision-Oriented Governance Under Uncertainty

The same detector can be useful for triage and unsuitable for removal. Let d be an action, H_π the policy event, and z severity, affected population, reversibility, and operational state. A local decision model is

$$d^*(o) = \arg \min_d \mathbb{E}[L(d, H_\pi, z) \mid o]. \quad (12)$$

The loss is a policy-dependent modeling choice, not evidence that all harms share a natural cardinal scale. Figure 7 illustrates two separate constructions; only panel (a) minimizes the four stated action losses.

9.1 From Discrimination to Actionable Probability

Accuracy, F1, and AUC measure discrimination, not actionable probabilities. Neural models can be miscalibrated under shift [21, 44, 107]; fitting a calibrator differs from validating residual calibration error [67]. Free-form uncertainty should respect semantic equivalence [66], but consistent harmful responses can show little variation while benign completions vary widely. Disagreement can prompt inspection; confidence still requires validation against the retained policy event. Further generations are worthwhile only when improved decisions justify their cost and additional harmful exposure under the operational budget; reduced uncertainty alone is insufficient.

Calibrate the actual event—an output, a session, or activation within a budget. Global, subgroup, post-selection, and new-version calibration are not equivalent. C3 shows why an output probability cannot simply be reused for repeated exposure. Global calibration can conceal identity-, language-, topic-, style-, or channel-specific error [1, 33, 129]. Report policy and deployment slices with uncertainty intervals.

9.2 Choosing Actions Under Costs and Review Capacity

Panels (b)–(c) of Figure 7 use a heuristic review band around the allow–block threshold. Their conditional automated loss excludes reviewer cost, error, and delay; it is not the optimized total loss from panel (a). With review fraction q , total loss would combine $(1 - q)L_{\text{auto}} + qL_{\text{human}}$ and operating/delay costs. The component losses must be conditional

Table 10. Proposed evidence requirements, safeguards, and monitoring for moderation actions; thresholds depend on the deployment.

Action	Proposed evidence before use	Proposed safeguard	Monitoring signal
Allow/annotate	Low risk in tested channel; benign controls	Context/provenance when relevant	Incidents and subgroup misses
Warn / regenerate	Moderate or sampling-sensitive risk	Policy reason and reformulation path	Overrides, retries, utility loss
Refuse/block	Strong match, severity, low threshold error	Appeal/exception path	Appeals, drift, bypass
Human review	Material uncertainty/disagreement	Context, worker support, escalation	Delay, disagreement, overturn
Restrict component	Static concern plus behavior tests	Quarantine, sandbox, expiring attestation	Bases, merges, triggers, incidents
Monitor only	Plausible but insufficient evidence	Logged rationale and expiry	Risk or severity shift

on the actual automated and reviewed subsets. Changing routing changes both subsets and their losses, not just the mixture weights; increasing review capacity therefore need not reduce the total.

Costs are asymmetric: a violent-threat miss differs from profanity, while blocking counterspeech or crisis support is harmful. XSTest and MMSafeAware expose over-sensitivity alongside misses; ModelCitizens finds systematic disagreement between community members and outside annotators [121, 140, 152]. Policy should preserve substantive disagreement rather than average it away. Review capacity and worker exposure further constrain routing, while automatic blocks add appeal and reversibility costs. These consequences explain why the same probability can support different actions across deployments. Table 10 specifies proposed safeguards, not corpus-validated universal thresholds.

Abstention can withhold a decision, request context, or initiate review, with different costs and information requirements. Selective classification formalizes risk-coverage; deferral/triage methods optimize the detector-reviewer system [36, 92, 101, 104]. Route cases where reviewers add reliable information: uncertainty alone is insufficient when reviewers also lack context or language expertise. Compare review with automation on the same routed cases, reporting reviewer error, waiting time, and unequal burdens across languages and groups. Removing difficult cases can improve unreviewed accuracy without improving joint performance.

Review packets need policy, context, version, evidence, and escalation rationale. Explanations aid diagnosis without validating decisions; appeals add evidence [93, 124]. For components, quarantine preserves an unresolved state and sandboxing tests activation. Static signals can prioritize investigation; release requires behavioral evidence for the intended composition.

9.3 Longitudinal Monitoring and Evidence Records

Model cards and datasheets document systems and data [34, 98]. Governance also needs explicit problem formulation, stakeholder meaning, and measurement validity [58, 109, 135], embedded in an institutional audit process [119]. Practitioner, checklist, impact-assessment, and lifecycle studies identify data-access, responsibility, incentive, and stakeholder barriers [49, 91, 97, 128, 139]. Records therefore need accountable policy owners, review leads, and system/component maintainers able to approve actions, handle exceptions, and trigger retests. An incident log becomes operational evidence only when responsibility and authority connect its findings to a response.

Appeals reveal selected false positives; allowed content without complaints remains largely unlabeled. Audit allowed, blocked, and deferred cases with recorded selection probabilities and access safeguards, preserving exposure denominators and policy versions. Fewer incidents after stricter blocking need not mean better detection; auditing need not release harmful content. Responses to prevalence, semantic, system, and adversarial drift overlap: version changes may jointly require annotation, recalibration, regression tests, and renewed attack search. Material coordinate changes require new evidence or narrower reuse, with retests addressing the affected inference.

10 Cross-Cutting Synthesis and Research Agenda

Changing what a system can do often changes what evaluation must observe. The synthesis connects this problem to the research questions, then separates evidence-appraisal tests from operational-safety tests.

10.1 Cross-Domain Findings and Research Questions

RQ1 concerns the event being measured. Labels depend on policy, context, annotators, and aggregation [18, 123, 131]. Artifact classification, dialogue events, and sampled-generator risk are different estimands; Tables 4–6 preserve their denominators. RQ2 then asks whether the evaluated object exposes the information needed for that event. Dialogue escalation, cross-modal linkage, prompt recovery, and triggered composition add information or access not observed by an artifact classifier [89, 127, 143, 155]. C1 and C4 locate the resulting gaps without treating generic backdoor success as toxicity.

For RQ3, policy, observation, sampling, versions, and budgets change score meaning [35, 62, 94]. C2 and C3 isolate label and event/regime substitution; larger benchmark averages cannot fill gaps concerning different claims. RQ4 therefore requires bounded claims specifying policy, access, observation, regime, action, and uncertainty. Functional tests, source-specific audits, and task-specific labels have empirical precedents; their integration into release and monitoring workflows requires separate validation.

Figure 8 is an organizational map of all 183 cited works, not a count of tested bridges. Content tests and bias measures offer task-specific diagnostics [6, 124]; guards and visual judgments require their own policies and calibration [3, 37, 55]; component inspection and activation studies leave release transport unresolved [80, 89, 163]. These assessments are based on the reported study designs, endpoints, and limitations.

Policy-conditioned measurement is the premise linking these findings. A detector can reproduce benchmark labels yet disagree with the governed population [33, 129, 140]; guard categories still require explicit crosswalks [3]. Label agreement and deployment-policy agreement are different comparisons. The three recurring mechanisms below concern evidence transfer after specifying and justifying the retained event.

Channel substitution links failures that otherwise appear modality-specific. History, visual instructions, cross-modal linkage, recovered concepts, and loaded adapters provide distinct routes to failure [41, 89, 127, 143, 155]. Each can expose behavior unobserved through another route. This motivates matched comparisons with and without the relevant access while retaining the outcome policy, rather than claims of ecosystem-wide migration or a common effect size.

Repeated-access amplification and judge dependence affect different links in that comparison. Inputs, seeds, versions, and budgets define generative events [35, 62, 94]: repeated opportunities can change event probability even when the single-output distribution is fixed. Judge dependence instead concerns whether observed outcomes are correctly identified. When a shared judge shapes data, intervention, and evaluation, agreement can masquerade as safety. Increasing sample size cannot resolve shared labeling error, just as independent adjudication cannot repair an unspecified access budget. A validated violation within the declared domain falsifies a zero-violation guarantee and establishes

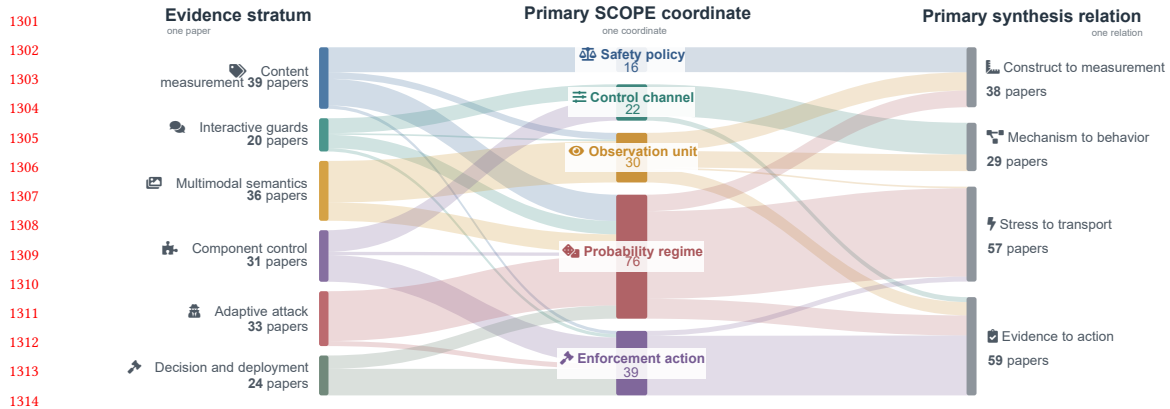


Fig. 8. Descriptive synthesis pathways across the 183 cited works. Each work is assigned one primary path linking an evidence stratum, a SCOPE coordinate, and a synthesis relation. Ribbon widths count publications and do not indicate evidence strength or successful transport.

reachability; it does not alone refute an arbitrary nonzero probability bound. With no observed failure, conclusions remain conditional on the sampled inputs, search procedure, and threat model.

10.2 Testing the Framework and Prioritizing Research

The five failure classes in Section 3 locate unsupported inferences; the mechanisms help account for recurring gaps without forming a complete causal model or a one-to-one mapping. A proposed assurance loop connects six stages: specify policy and targets; inventory channels and carriers; acquire direct and tested transport evidence; validate judges and unresolved cases; choose actions under calibration, severity, reversibility, and capacity; and monitor incidents, appeals, drift, and expiry. Owners and retest triggers connect unresolved assumptions to narrower claims or new comparisons.

A proposed test would compare SCOPE with an ordinary reporting checklist, giving independent analysts identical evidence and target claims. Include supported narrow claims and unsupported extensions, missing reporting and demonstrated failures, and single- and multiple-coordinate changes. Policy-consistent counterexamples should test correct gap identification beyond analyst agreement or indiscriminate rejection. Measure agreement on unsupported extensions, justification traceability, follow-up-test relevance, and analysis effort. This tests evidence appraisal: whether analysts locate relevant gaps and request proportionate tests. More cautious language, longer records, or completed fields do not demonstrate improved system safety.

Table 11 prioritizes artifacts by evidence dependency: unreliable event definitions undermine transport comparisons, and unvalidated transport weakens release claims. Measurement, composition, and deployment studies may iterate; the ordering identifies assumptions needed for subsequent inferences, not a fixed timetable.

Near-term crosswalks must distinguish translation invariance from community-specific judgments [122, 140]. Component attestations must then bind static signals to executed bases, triggers, utility, and expiry [80, 89, 163]. Deployment trials address a different endpoint from the analyst comparison above: whether the resulting decisions reduce total loss and unequal burdens under a stated policy and exposure regime. Better gap identification or documentation is insufficient evidence of that operational benefit.

Table 11. Proposed research priorities arising from gaps in governance validity.

Priority	Proposed artifact	Evaluation criterion	Invalid inference prevented
Policy crosswalks with disagreement	Versioned labels, rules, targets, contexts, annotator groups	Predict held-out policy/community decisions and disagreement	Binary toxicity transfers universally
Channel-explicit manifests	Machine-readable prompt, history, modality, retrieval, tool, component access	Residual risk per channel/composition	One interface certifies the system
Sequential estimation	Sampling/multi-turn protocols, budgets, stopping, intervals	Event and best-of- n stability across versions	One output measures risk
Independent judge audits	Policy-grounded judges, human subset, indeterminate label	Judge error/dependence and group calibration	Automatic evaluator is ground truth
Component attestation	Signed metadata, static scan, sandbox, compatible bases, expiry	Retest after merge, update, or trigger search	Clean file implies safe composition
Deployment trials	Threshold, risk-coverage, review budget, appeals, incidents	Utility, subgroup burden, exposure, overturns, severe misses	Higher AUC implies better governance

10.3 Limits of Unification

Unifying evidence conditions does not supply a common policy or severity scale. The non-exhaustive corpus and the review limitations constrain the coverage of the proposed synthesis; the worked profiles demonstrate a coding procedure, not inter-rater reliability or validation of SCOPE. Coordinates overlap, direct support can be weak, and a successful bridge for one change does not validate joint changes. Heterogeneous endpoints preclude a pooled effect, while normative disagreement may persist after measurement improves. Governance validity can expose the evidence required by a chosen policy, but cannot settle that policy’s legitimacy or certify future robustness.

11 Conclusion

Toxicity detection concerns policy-conditioned events, not interchangeable scores. Moving from artifacts to conversations, visual generations, or reusable components changes what must be observed and tested. SCOPE makes those changes explicit while keeping support status separate from evidence quality. Its practical output is a bounded claim: what the evidence establishes, which action it can inform, and which comparison is still needed. Component composition and longitudinal deployment remain important gaps; an assurance loop is a proposal for testing and maintaining these claims, not a certificate of safety.

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